

Relational Schema

1. Transport

Transport(VehicleID:number(10), ManufactureYear:number(4), Capacity: number(4), WheelchairAccessibility: char(1))

PK: Vehicle ID

CK: None

FK: None

2. Bus

Bus(VehicleID: number(10), LiscencePlate: char(10), HomeGarage: char(60), FuelType: char(25))

PK: VehicleID

CK: LiscencePlate

FK: VehicleID references Transport(VehicleID)

3. SubwayTrain

SubwayTrain(VehicleID: number(10), TrainModel: char(40), NumberOfCars: number(2), TrainSetNumber: char(20))

PK: VehicleID

CK: TrainSetNumber

FK: VehicleID references Transport(VehicleID)

4. Passenger

Passenger(PassengerID: number(10), FirstName: char(20), LastName: char(20), PassengerCategory: char(20))

PK: PassengerID

CK: None

FK: None

5. Passes

Passes(TicketID: number(10), PassengerID: number(10), TravellingStatus: char(20), AmountPaid: number (6))

PK: TicketID

CK: None

FK: PassengerID references Passenger(PassengerID)

6. ZonePass

ZonePass(TicketID: number(10), NumberOfValidZones: number(2))

PK: TicketID

CK: None

FK: TicketID references Passes(TicketID)

7. TimedPass

TimedPass(TicketID: number(10), StartTime: number(4), EndTime: number(4), PassType: char(30))

PK: TicketID

CK: None

FK: TicketID references Passes(TicketID)

8. Route

Route(RouteID: number(10), RouteName: char(60), Direction: varchar(20), Description: char(200), Colour: char(20))

PK: RouteID

CK: None

FK: None

9. Schedule

Schedule(RouteID: number(10), StartTime: number(4), Endtime: number(4)), Frequency: number(4), DayType: char(20))

PK: (RouteID, StartTime)

CK: None

FK: RouteID references Route(RouteID)

10. Zone

Zone(ZoneNumber: number(3), Name: char(50), Colour: char(20), Description: char(200))

PK: ZoneNumber

CK: None

FK: None

11. TransportStation

TransportStation(StationID: number(10), ZoneNumber: number(3), StationName: char(80), Address: char(150), WheelchairAccessibility: char(1))

PK: StationID

CK: None

FK: Zone Number references Zone(ZoneNumber)

12. RouteStop

RouteStop(RouteID: number(10), StopNumber(4), StationID: number(10))

PK: (RouteID, StopNumber)

CK: None

FK: RouteID references Route(RouteID), StationID references TransportStation(StationID)

13. Gate

Gate(GateID: number(10), StationID: number(10), ActivationStatus: char(20), EnterOrExit: char(10), TotalEnterExitCount: number(10))

PK: GateID

CK: None

FK: StationID references TransportStation(StationID)

Relations from Many-to-many Relationships

14. RunsOn

RunsOn(VehicleID: number(10), RouteID: number(10))

PK: (VehicleID, RouteID)

CK: None other than primary key

FK: VehicleID references Transport(VehicleID), RouteID references Route(RouteID)

15. ValidEntranceAt

ValidEntranceAt(TicketID: number(10), ZoneNumber: number(3))

PK: (TicketID, ZoneNumber)

CK: None other than pk

FK: TicketID references ZonePass(TicketID), ZoneNumber references Zone(ZoneNumber)

16. ValidExitAt

ValidExitAt(TicketID: number(10), ZoneNumber: number(3))

PK: (TicketID, ZoneNumber)

CK: None other than pk

FK: TicketID references ZonePass(TicketID), ZoneNumber references Zone(ZoneNumber)

17. PassEnter

PassEnter(TicketID: number(10), GateID: number(10))

PK: (TicketID, GateID)

CK: none other than pk

FK: TicketID references Passes(TicketID), GateID references Gate(GateID)

18. PassExit

PassExit(TicketID: number(10), GateID: number(10))

PK: (TicketID, GateID)

CK: none other than pk

FK: TicketID references Passes(TicketID), GateID references Gate(GateID)

Functional Dependencies

1. Transport

VehicleID -> ManufactureYear, Capacity, WheelchairAccessibility

2. Bus

VehicleID -> LicencePlate, HomeGarage, FuelType

LicencePlate -> VehicleID, HomeGarage, FuelType

3. SubwayTrain

VehicleID -> TrainModel, ~~NumberOfCars~~, TrainSetNumber, Manufacturer

TrainSetNumber -> VehicleID, TrainModel, ~~NumberOfCars~~, Manufacturer

In the database, every subway train model has a fixed number of cars.

4. Passenger

PassengerID -> FirstName, LastName, PassengerCategory

5. Passes

TicketID -> PassengerID, TravellingStatus, AmountPaid

6. ZonePass

TicketID -> NumberOfValidZones

7. TimedPass

TicketID -> StartTime, EndTime, PassType

8. Route

RouteID -> RouteName, Direction, Description, Colour

9. Schedule

RouteID, StartTime -> Frequency, DayType, EndTime

10. Zone

ZoneNumber -> Name, Colour, Description

11. TransportStation

StationID -> ZoneNumber, StationName, WheelchairAccessibility, Address

12. RouteStop

RouteID, StopNumber -> StationID

13. Gate

GateID -> StationID, ActivationStatus, EnterOrExit, TotalEnterExitCount

RunsOn, ValidEntranceAt, ValidExitAt, PassEnter, and PassExit are relations that contain only key attributes.